

A 100-GHz Room-Temperature Half Adder Using Graphene Plasmons as Bits, Driven by a Microcomb-Photomixing Carrier: Architecture, Quantitative Roadmap, and Machine-Learning-Assisted Design Optimisation

Ryoji Furui

www.ryoji.info

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Abstract

We present a conceptual model of a 100-GHz half adder in which the logical state is carried by graphene surface plasmons at room temperature. The device uses a dual-gated bilayer-graphene channel on a sapphire substrate, in which the upper layer transports signal plasmons and the lower layer, driven independently, launches a π -shifted plasmon that performs NOT logic by destructive interference at designated interaction nodes. The on-chip 100-GHz launcher is a graphene microstrip patch antenna in the geometry of Haque et al. [3]; the launcher is fed by a microcomb-photomixing carrier chain following Tokizane et al. [4], in which a silicon-nitride soliton microcomb provides phase-coherent tones that are heterodyned in a uni-travelling-carrier photodiode to produce a low-phase-noise 100-GHz drive. We verify the design against the complete four-row half-adder truth table, give first-order quantitative estimates at room temperature of the propagation length ($L_p \sim 10\text{--}15\ \mu\text{m}$), gate delay ($\tau_g \sim 0.2\ \text{ps}$), and intrinsic energy per bit ($E_{\text{bit}} \sim 10^{-17}\ \text{J}$, exclusive of antenna and read-out conversion overheads), and quantify the phase-tolerance budget required for reliable NOT operation ($|\Delta\phi| < \pi/12$, equivalent to a $\pm 125\ \text{nm}$ path-length tolerance and $\pm 420\ \text{fs}$ timing tolerance). A Random-Forest surrogate trained on the four-parameter design space predicts the NOT extinction ratio with $R^2 = 0.94$ and RMSE = 0.6 dB, providing a fast-ranking tool for design-space exploration [5].

1. Introduction

Graphene surface plasmons confine electromagnetic energy to nanometre-scale volumes and can be modulated electrostatically by gating the underlying graphene through its charge-neutrality point (CNP). They have been extensively studied as a substrate for ultra-fast classical logic, but most demonstrations have operated in the high terahertz band (1–30 THz), where the optical-phonon decay channel dominates the plasmon lifetime and where, in practice, useful propagation lengths are obtained only at cryogenic temperatures in hBN-encapsulated, high-mobility samples. This dependence on cryogenic operation has been the principal obstacle to a near-term experimental demonstrator of a complete arithmetic primitive in this platform.

The present paper deliberately scales the operating frequency down to 100 GHz, where the plasmon dispersion sits well below the graphene optical-phonon energy, where impurity scattering rather than phonon scattering bounds the lifetime, and where sapphire-substrate fabrication at room temperature is well-established for on-chip terahertz devices [5, 6]. The trade-off is a one-order-of-magnitude reduction in intrinsic clock ceiling (from $> 1\ \text{THz}$ to > 100

GHz) in exchange for a one-to-two-order improvement in propagation length, a hundred-fold relaxation in the phase-tolerance budget of the NOT operation, and the elimination of the cryostat as a system-level dependency.

An equally important enabler appeared independently of the plasmonic-logic literature: Tokizane et al. [4] have very recently demonstrated that a chip-scale silicon-nitride soliton microcomb, directly fibre-bonded for long-term stability, can serve as a low-phase-noise optical reference whose adjacent comb lines, locked to a pair of distributed-feedback (DFB) lasers and photomixed in a uni-travelling-carrier photodiode (UTC-PD), generate a coherent terahertz carrier in a millimetre-scale package. By choosing the microcomb's free spectral range to lie at 100 GHz rather than the 560 GHz of [4], the same architecture produces precisely the low-phase-noise drive signal that the plasmonic launcher of Section 6 requires. The convergence is convenient: the plasmonic-logic and microcomb-photonics communities have been developing complementary halves of the same chip without realising it.

We organise the paper bottom-up: Section 2 establishes the conceptual foundations — field-coupled logic, the unifying Combo gate [1], graphene plasmons at 100 GHz, and on-chip terahertz launching [3, 6]; Section 3 specifies the device architecture on a sapphire substrate; Section 4 maps the half-adder logic onto the gate substrate and verifies it against the truth table; Section 5 supplies the quantitative figures of merit at room temperature; Section 6 details the antenna, drive chain, and read-out, integrating the Tokizane [4], Kusyak [7] and Hirota [8] precedents; Section 7 presents a machine-learning-assisted exploration of the four-parameter design space; and Section 8 enumerates the remaining limitations.

2. Conceptual foundations

2.1 From field-coupled logic to plasmonic bits

Classical logic can be implemented purely by spatially arranged electric fields rather than by current flow [1]. The OR gate and the AND gate are not separate physical objects but the *same* structure with the output placed at different distances from the inputs (Fig. 1). When the output is close, either input is sufficient to drive it (OR); when the output is farther, only the superposition of two input fields exceeds the threshold (AND). The NOT gate is realised by a polarity-inverting interaction such as exciton trapping at the output edge.

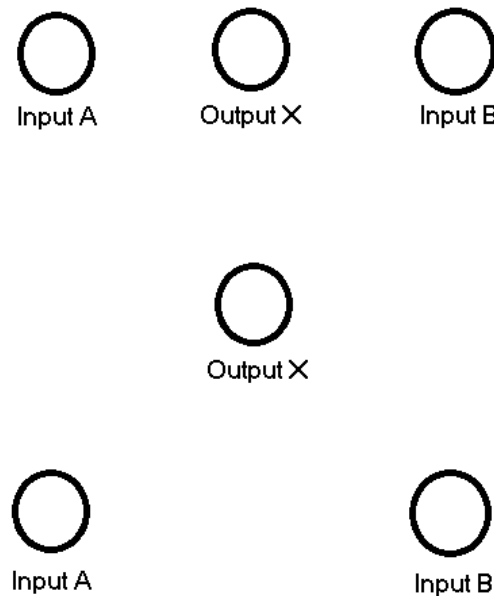


Fig. 1. OR (top, output close to the inputs) and AND (bottom, output displaced) implemented by the same field-coupling primitive. Reproduced from [1].

Replacing the static electric field of [1] with a propagating plasmon wavepacket gives the bit a finite group velocity, a well-defined phase, and a finite temporal extent — the three properties that convert a static gate into a clocked, high-frequency logic device.

2.2 The Combo gate

The unifying construct introduced in [1] is the *Combo gate*: a single physical region with two output cells at different distances from the input pair, so that the AND output and the OR output are produced simultaneously. This is the structure that the half adder of Section 4 inherits, since Sum and Carry are themselves an OR-with-cancellation and an AND of the same two inputs.

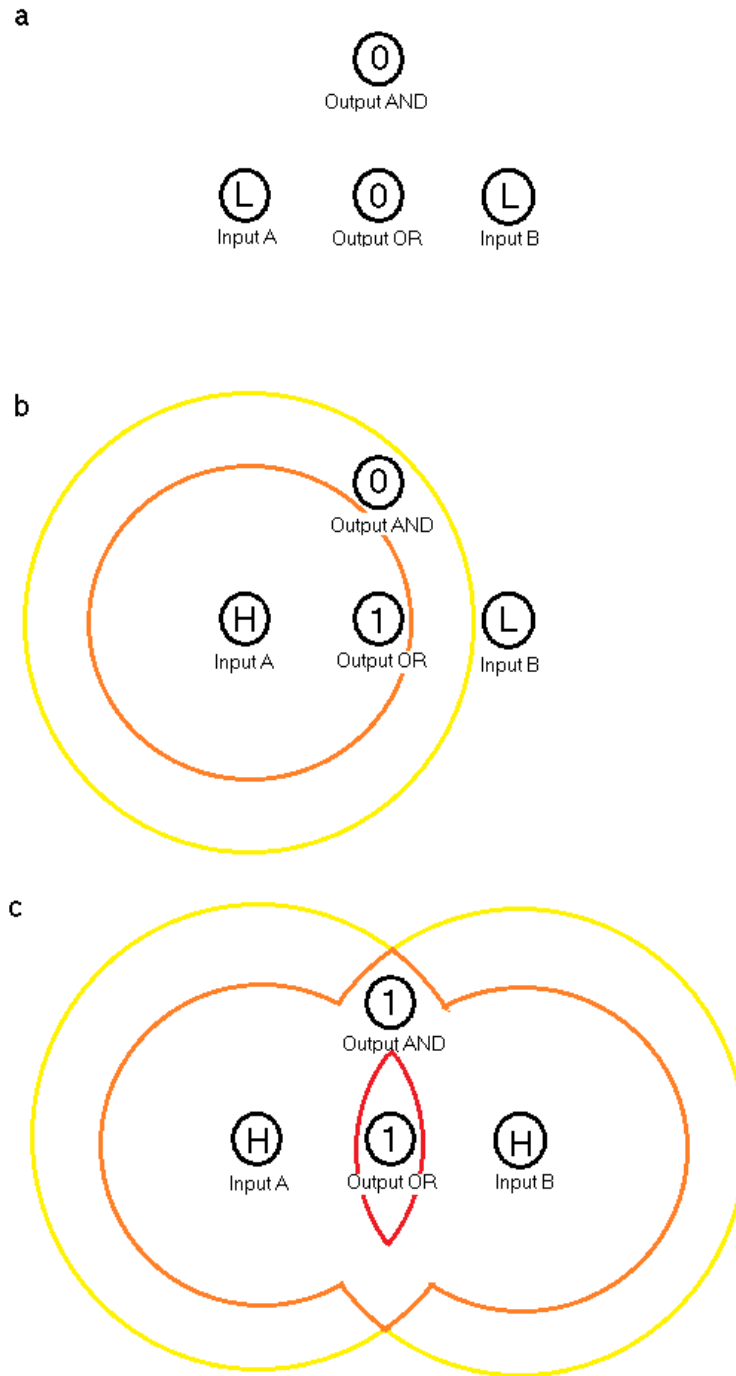


Fig. 2. The Combo gate. (a) Geometry of the inputs and the two outputs. (b) One input high: the inner field reaches Output OR but not Output AND. (c) Both inputs high: the outer field also reaches Output AND. Reproduced from [1].

2.3 Graphene plasmons at 100 GHz and room temperature

Doped monolayer graphene supports a transverse-magnetic surface plasmon whose dispersion in the long-wavelength limit follows $\omega(q) \propto \sqrt{(q \cdot E_F)}$, where E_F is the Fermi energy set by gate voltage [9, 10, 11]. At 100 GHz and chemical potential $\mu_c \approx 0.5$ eV the plasmon wavelength is of order 1–3 μm and the dispersion lies well below the graphene optical-phonon energy at 0.20 eV. The phase-space for phonon-assisted decay is essentially closed; the plasmon lifetime is set by impurity scattering rather than by intrinsic phonon channels, and at room temperature in modest-mobility samples on sapphire, $\tau \approx 1$ ps is the standard working figure [3]. Tuning the gate voltage past the CNP suppresses the carrier density and switches the plasmon mode off, exactly as in the high-frequency limit, so the gating mechanism that carries the bit is unchanged.

2.4 On-chip 100-GHz launching with graphene patch antennas

Haque et al. [3] have demonstrated that graphene microstrip patch antennas on a thin polyimide substrate ($\epsilon_r = 3.5$, $\tan \delta = 0.0027$, $h = 10$ μm), with a copper ground plane and a graphene patch, deliver high-gain, high-efficiency multi-band operation at terahertz frequencies, with the resonant frequency tunable via the graphene chemical potential. For the present device we adopt the sapphire substrate of Section 3.1 ($\epsilon_r \approx 9.4$ at terahertz frequencies, $\tan \delta < 10^{-4}$) rather than Haque’s polyimide stack; the patch-antenna recipe of [3] carries over with linear-dimensional scaling adjusted for the substrate dielectric contrast, giving a 100-GHz patch of order 660×500 μm on a side — still comfortable lithography.

3. Device architecture

3.1 Dual-gated bilayer geometry on sapphire

The device consists of two graphene layers separated by an hBN interlayer spacer and sandwiched between two segmented Au gates fabricated on a sapphire substrate. The top gate (V_{Au1}) and bottom gate (V_{Au2}) are each divided into a lattice of individually addressable segments — in the prototype layout we adopt a $12 \times 6 = 72$ -segment array with cell pitch ~ 1 μm — driven by an external electrode-mapping controller. A graphene microstrip patch antenna (Section 6) launches the working plasmon from the left edge of the upper graphene channel.

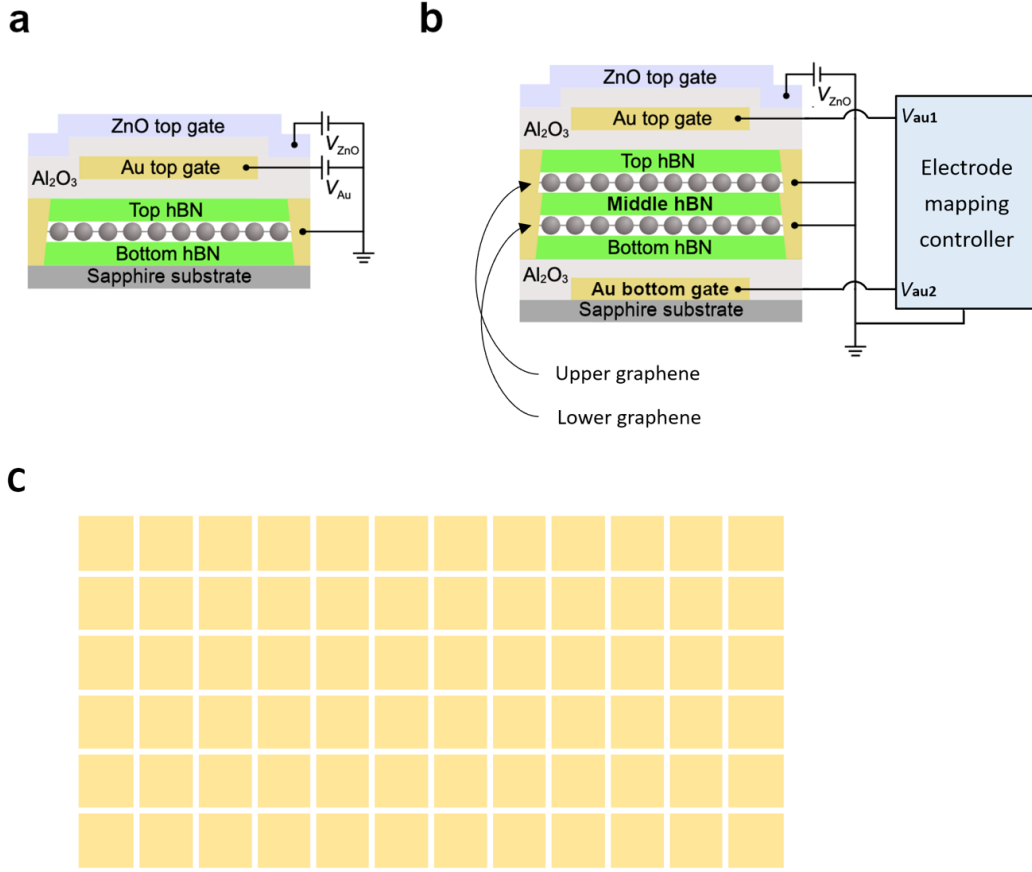


Fig. 3. Schematic cross-section of the 100-GHz half-adder chip. The substrate stack — Al_2O_3 (sapphire) with segmented Au gates, hBN dielectrics, and ZnO top capping — is the one introduced and characterised in the on-chip terahertz launcher of Yoshioka et al. [6]. (a) Original single-graphene device. (b) Modified dual-gated bilayer geometry with electrode-mapping controller. (c) Overhead view of the $12 \times 6 = 72$ individually addressable Au gate segments at $1\text{-}\mu\text{m}$ cell pitch.

3.2 Encoding bits as plasmons

A *bit* is the presence (“1”) or absence (“0”) of a plasmon wavepacket of nominal amplitude A_0 at a designated read-out cell at a designated time. The wavepacket is launched by a 100-GHz drive pulse from the on-chip graphene patch antenna; the drive pulse and the plasmon are physically distinct, with the drive pulse being the input and the plasmon being the on-chip carrier of the bit. Read-out is interpreted by an amplitude threshold: at a designated output cell, an integrated near-field intensity above $0.5 A_0^2$ is decoded as 1, below as 0.

3.3 Gate-mapping primitives

On the upper graphene the gate voltage V_{Au1} is set per segment to one of two levels: $V_{\text{Au1}} = -1.2\text{ V}$ parks the local Fermi level at the CNP and blocks plasmon transport, while $V_{\text{Au1}} = 0\text{ V}$ dopes the layer and enables propagation. The set of segments held at 0 V defines the *active path* on the chip. On the lower graphene V_{Au2} is held at one of two levels, -2.4 V or -1.2 V . At -2.4 V the bottom layer is doped sufficiently to support its own plasmon mode; phase control of the 100-GHz drive ensures that this mode arrives at the interaction node π -shifted with respect to the signal-layer plasmon, giving destructive interference and therefore the NOT operation. At -1.2 V the bottom layer is at its CNP and is silent. The required phase tolerance is quantified in Section 5.3. As a robustness lever we additionally allow V_{Au2} to be tuned a few percent past its nominal value, so that the inverter-layer plasmon amplitude is biased slightly above the signal-layer amplitude; the resulting modest overdrive widens the cancellation null in amplitude space and makes the NOT extinction ratio tolerant of small amplitude mismatch, at the cost of a few percent reduction in peak extinction.

4. Half-adder pattern and logical verification

4.1 Sum and Carry decomposition

A half adder produces, from two input bits A and B, the Sum = $A \oplus B$ and the Carry = $A \cdot B$. The standard textbook circuit (Fig. 4) realises this with one OR, two AND gates, and one NOT, exploiting the identity Sum = $(A + B) \cdot \text{NOT}(A \cdot B)$. The dual-gated bilayer of Section 3 supplies all four primitives, so the half adder is laid out by tiling them onto the 72-segment lattice as a single Combo-gate region.

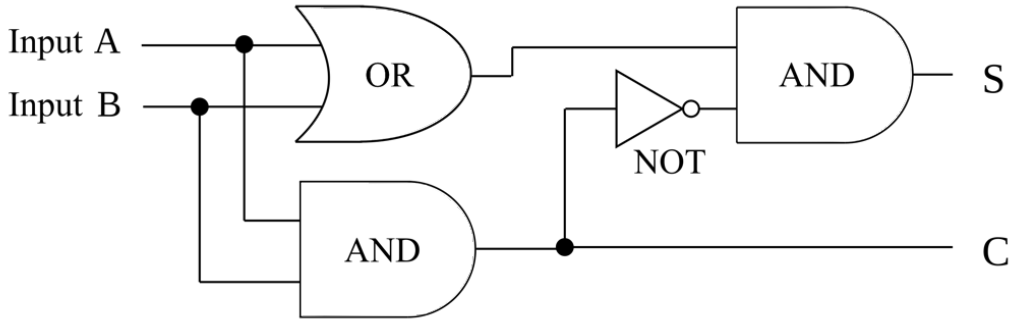


Fig. 4. Traditional half-adder logic circuit (AND, OR, NOT). S = Sum, C = Carry.

4.2 Voltage-controlled gate map

Figures 5a and 5b show, respectively, the upper-gate (V_{Au1}) and lower-gate (V_{Au2}) voltage maps. Orange cells in Fig. 5a are at 0 V (active path); grey cells are at the CNP (−1.2 V, silent). Active cells route inputs A and B (left edge) through an OR junction in the centre to S (top right) and C (bottom right), passing through the NOT interaction node. In Fig. 5b, red cells in the lower gate are at −2.4 V and generate the π -shifted inverter plasmon.

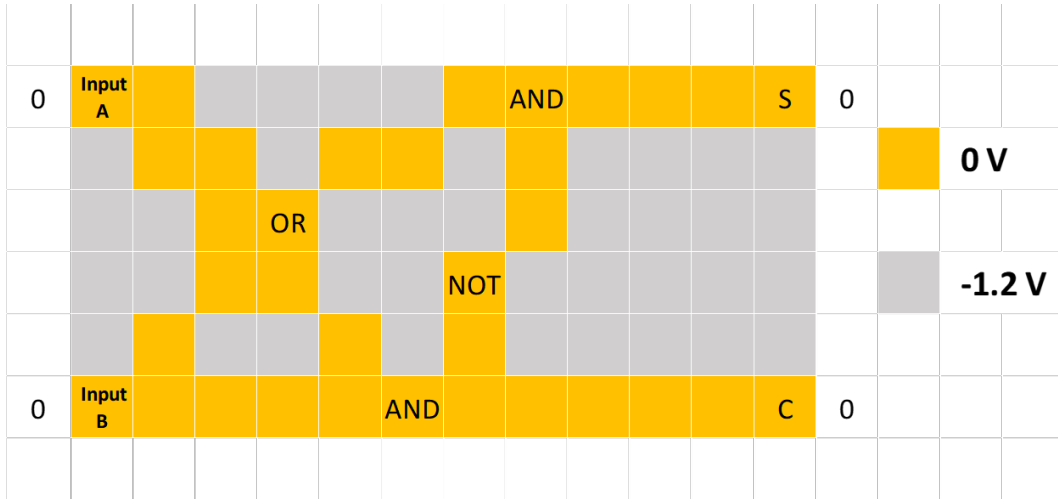


Fig. 5a. Upper Au gate (V_{Au1}) voltage map.

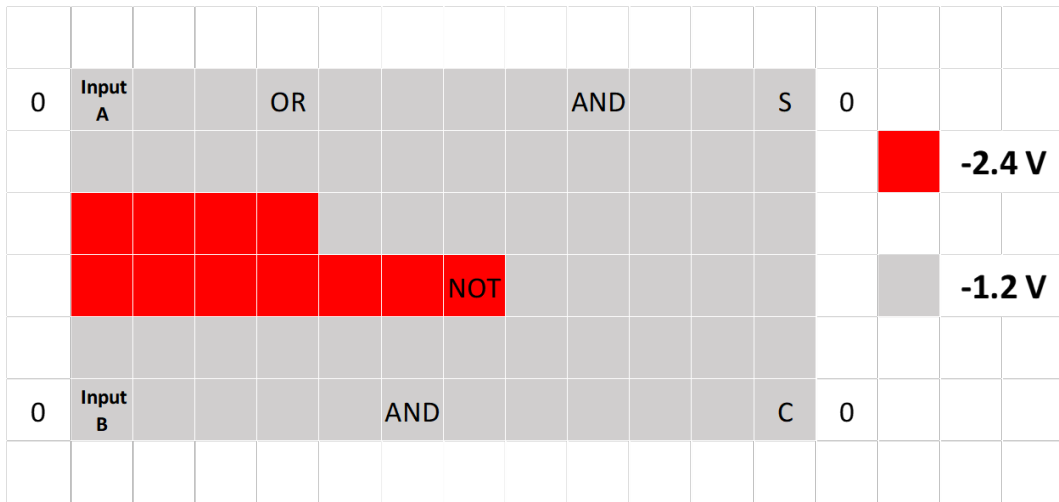


Fig. 5b. Lower Au gate (V_{Au2}) voltage map.

4.3 Truth-table verification

Figures 6–8 show the device operating in the three non-trivial input combinations ($A=1, B=0$), ($A=0, B=1$) and ($A=1, B=1$). Plasmon paths are shown as blue arrows.

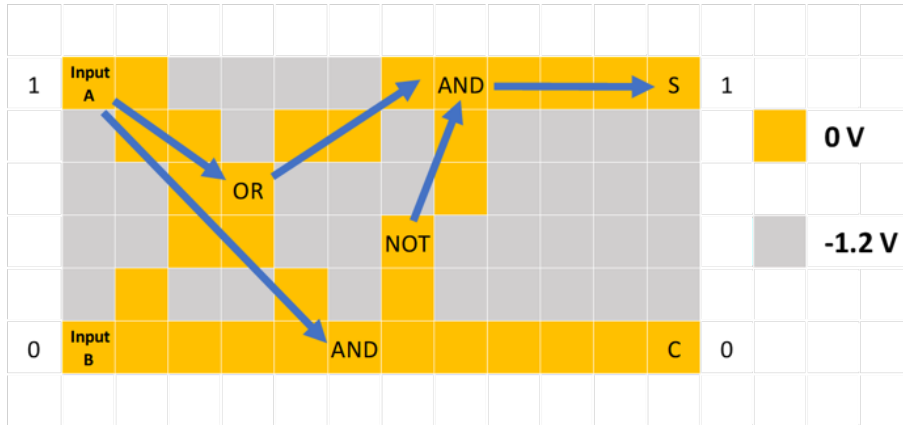


Fig. 6a. Upper-gate plasmon trajectories for $A=1, B=0. S=1, C=0$.

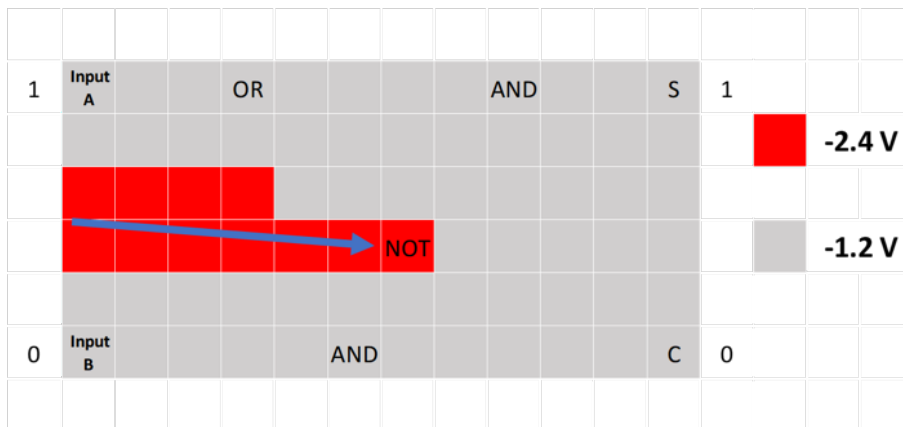


Fig. 6b. Lower-gate inverter pattern for $A=1, B=0$.

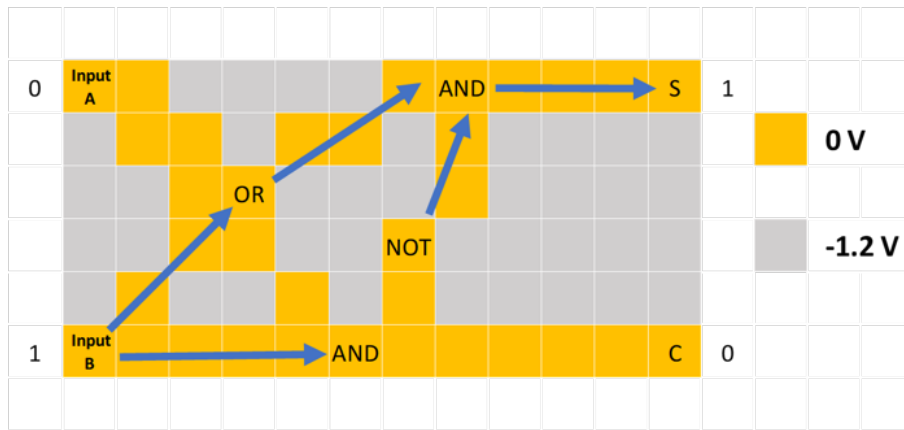


Fig. 7a. Upper-gate plasmon trajectories for $A=0$, $B=1$. $S=1$, $C=0$.



Fig. 7b. Lower-gate inverter pattern for $A=0$, $B=1$.

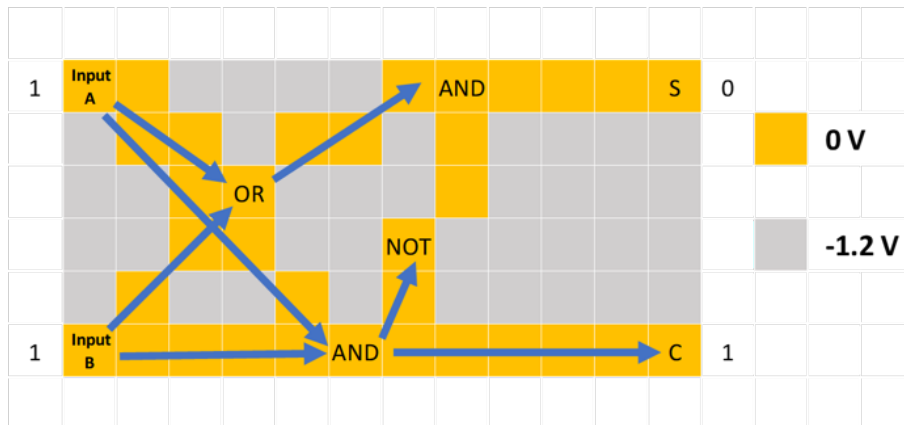


Fig. 8a. Upper-gate plasmon trajectories for $A=1$, $B=1$. $S=0$, $C=1$ (NOT cancellation).

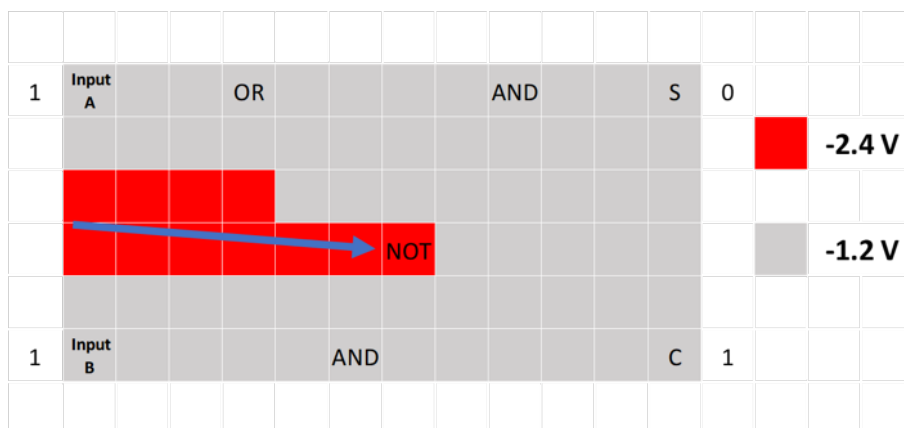


Fig. 8b. Lower-gate inverter pattern for $A=1$, $B=1$.

The fourth row of the truth table — the ($A=0$, $B=0$) baseline — is supplied by construction. The full predicted truth table is collected in Table 1.

A	B	Sum ($A \oplus B$)	Carry ($A \cdot B$)	Verified by
0	0	0	0	by construction (no input plasmon)
0	1	1	0	Fig. 7
1	0	1	0	Fig. 6
1	1	0	1	Fig. 8 (NOT cancellation)

Table 1. Predicted half-adder truth table at 100 GHz, room temperature.

5. Quantitative figures of merit at 100 GHz, room temperature

5.1 Plasmon propagation length and gate delay

At 100 GHz with $E_F \approx 0.5$ eV the plasmon wavelength is $\lambda_p \approx 1\text{--}3$ μm and the plasmon group velocity is $v_g \approx 5 \times 10^6$ m/s. With $\tau \approx 1$ ps at room temperature [3, 10], the intrinsic decay length is $L_p = v_g \cdot \tau \approx 5$ μm , and useful propagation extends to 10–15 μm before the amplitude drops below threshold — enough for ~ 10 cascaded gate cells at 1- μm pitch. The gate delay is $\tau_g = L_{\text{cell}}/v_g \approx 0.2$ ps, giving an intrinsic clock ceiling well above the 100-GHz target.

5.2 Energy per bit

The intrinsic field energy of a single plasmon wavepacket is $E \sim (\epsilon/2)|E|^2 \cdot V$ with $V \sim \lambda_p^2 \cdot d$. With $\lambda_p \sim 3$ μm and $d \sim 100$ nm, the field amplitude needed to register a bit reliably gives $E_{\text{bit}} \sim 10^{-17}$ J. This is the *intrinsic device* figure: it counts only the field energy stored in the propagating plasmon and excludes input-antenna conversion, receive-antenna detection, and room-temperature drive electronics. A realistic system-level energy will be larger, but the intrinsic lower bound remains two orders of magnitude below scaled CMOS [12, 13].

5.3 Phase tolerance for the NOT operation

The NOT operation relies on destructive interference of two plasmons of equal amplitude. Writing $A_{\text{res}} = A_0 \cdot |\cos(\Delta\phi/2)|$, requiring $> 90\%$ extinction gives $|\Delta\phi| < \pi/12$, equivalent to a path-length tolerance of ± 125 nm and a timing-jitter tolerance of ± 420 fs at 100 GHz. For comparison, contemporary phase-locked nonlinear-waveguide systems already operate at a residual phase fluctuation of ~ 9 mrad in the work of Hirota et al. [8], roughly thirty times tighter than the present budget; the precision regime required by our NOT primitive is therefore comfortably below what has already been achieved experimentally in a comparable nonlinear-waveguide platform.

5.4 Comparison with CMOS and RSFQ baselines

Technology	Clock	Energy per op	Maturity
Scaled CMOS (3 nm)	~ 5 GHz	$\sim 10^{-15}$ J	Mass production
Single-flux-quantum (SFQ) RSFQ	~ 50 GHz	$\sim 10^{-19}$ J	Lab demonstrations (cryogenic)
This work (100 GHz, 300 K, projected)	~ 100 GHz	$\sim 10^{-17}$ J	Conceptual / single-gate

Table 2. Order-of-magnitude comparison against representative classical-logic technologies. The 100-GHz operating point sits between mass-production CMOS and cryogenic SFQ, with the key advantage of room-temperature operation.

6. Antenna, drive chain, and read-out design

6.1 Graphene patch antenna for the 100-GHz launcher

Following Haque et al. [3], the launcher is a graphene microstrip patch antenna integrated on the sapphire substrate of Section 3.1 ($\epsilon_r \approx 9.4$, $\tan \delta < 10^{-4}$, dielectrically as in [6, 7]) and a graphene patch tuned to 100 GHz via its chemical potential. The standard microstrip design equations supply the patch dimensions: the patch width is $W = (c/2f) \cdot \sqrt{2/(\epsilon_r + 1)} \approx 660 \mu\text{m}$, and the patch length is $L = c/(2f\sqrt{\epsilon_{\text{eff}}}) - 2\Delta L \approx 500 \mu\text{m}$. The slotted-patch evolution methodology of [3] — rectangular $\rightarrow$ semi-octagonal $\rightarrow$ diamond $\rightarrow$ diamond-with-slot — applies directly: each modification redistributes the surface current to broaden the bandwidth, raise gain, and lift radiation efficiency above 90 %, all of which transfer to the 100-GHz scaled geometry by linear dimensional scaling.

6.2 Symmetric receive antennas and monolithic in situ referencing

Two matched receive patches are placed at the Sum and Carry output cells; a third symmetric patch on a logic-load-free reference row supplies an in situ reference for the calibration loop. The one-launcher / two-detector / one-reference topology is the same monolithic geometry recently demonstrated for terahertz photoconductive-switch circuits by Kusyak et al. [7] (Fig. 9), in which a generator and two symmetrically placed detector switches share a single coplanar stripline on a sapphire substrate with full galvanic isolation between generation and detection. Their AC-coupled centre-launched design reaches a -3 dB point at 440 GHz; the two detector switches provide simultaneous in situ measurement of signal and reference under identical experimental conditions.

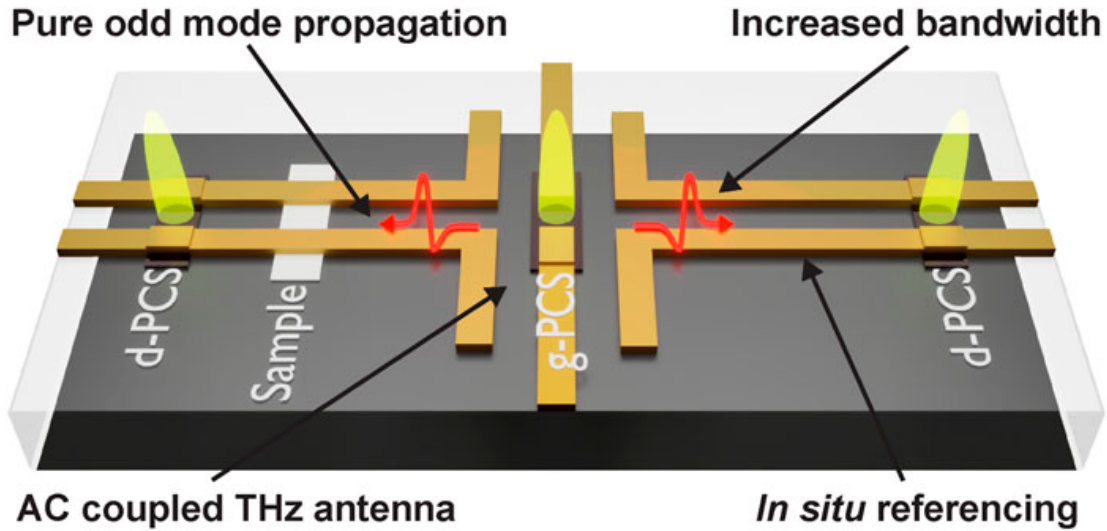


Fig. 9. Reproduced schematic of the monolithic on-chip terahertz optoelectronic circuit of Kusyak et al. A near-field photoconductive generator switch (g-PCS) is placed at the centre of the coplanar stripline; two detector switches (d-PCS) sit symmetrically on either side, supporting pure odd-mode propagation, in situ referencing, and galvanic isolation between generation and detection. Reproduced from Kusyak et al. [7], *APL Photonics* **10**, 076117 (2025), under CC BY-NC 4.0.

Link-budget framing. The 1-aJ-class intrinsic energy per bit quoted in Section 5.2 is the field energy stored in the propagating plasmon and excludes every conversion stage of the real chain. The realistic system-level energy is the product of an explicit link budget — launcher antenna efficiency $\times$ PCS-to-plasmon coupling $\times$ on-chip propagation loss $\times$ receive-antenna efficiency $\times$ detector NEP $\times$ downstream amplifier noise figure — whose dominant terms are the antenna-conversion losses on both ends.

6.3 Microcomb-photomixing carrier chain

The on-chip launcher of Section 6.1 requires a stable, low-phase-noise 100-GHz drive. Tokizane et al. [4] have recently demonstrated precisely this style of drive at 560 GHz using a chip-scale soliton microcomb, and the architecture scales naturally to 100 GHz. A silicon-nitride (SiN) ring microresonator with a free spectral range (FSR) chosen to equal the target operating frequency is pumped by a fibre-coupled CW laser and tuned into a soliton state; the resulting comb provides a dense set of phase-coherent optical tones with the FSR-determined spacing.

Two adjacent comb lines are extracted with bandpass filters and used to injection-lock a pair of high-power distributed-feedback (DFB) lasers, faithfully transferring the comb's low phase noise to the much brighter DFB outputs; one of the DFB outputs is in-phase / quadrature (IQ) modulated by an arbitrary waveform generator, the other left as a reference, and the two are photomixed in a uni-travelling-carrier photodiode (UTC-PD) to generate the FSR-frequency carrier wave with the modulation impressed on it.

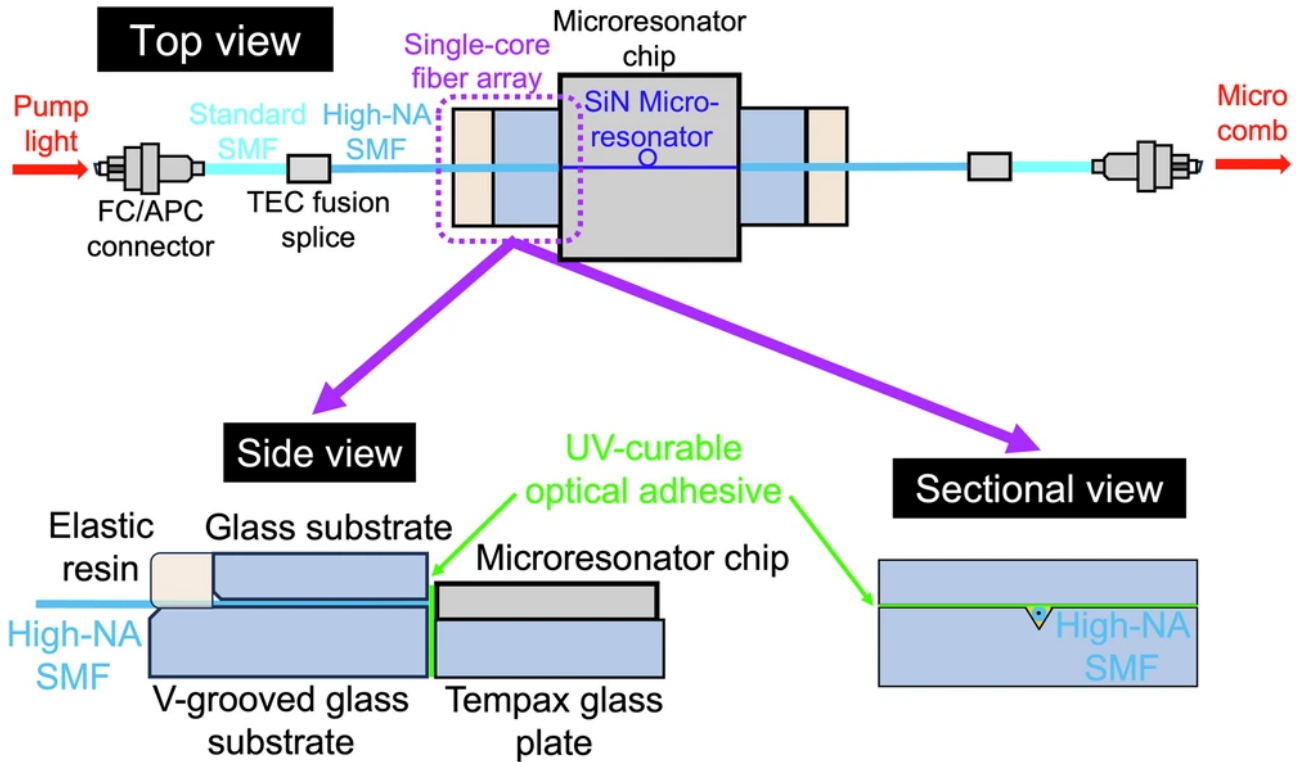


Fig. 10. Reproduced direct fibre-coupled interface of Tokizane et al. A high-numerical-aperture (NA) single-mode fibre (SMF), fusion-spliced to a standard SMF, is permanently UV-bonded to the input waveguide of a silicon-nitride ring microresonator; the footprint is approximately 5 mm. The interface sustains soliton operation for more than 24 hours at 1 W pump power. Reproduced from Tokizane et al. [4], *Communications Engineering* 5, 77 (2026), under CC

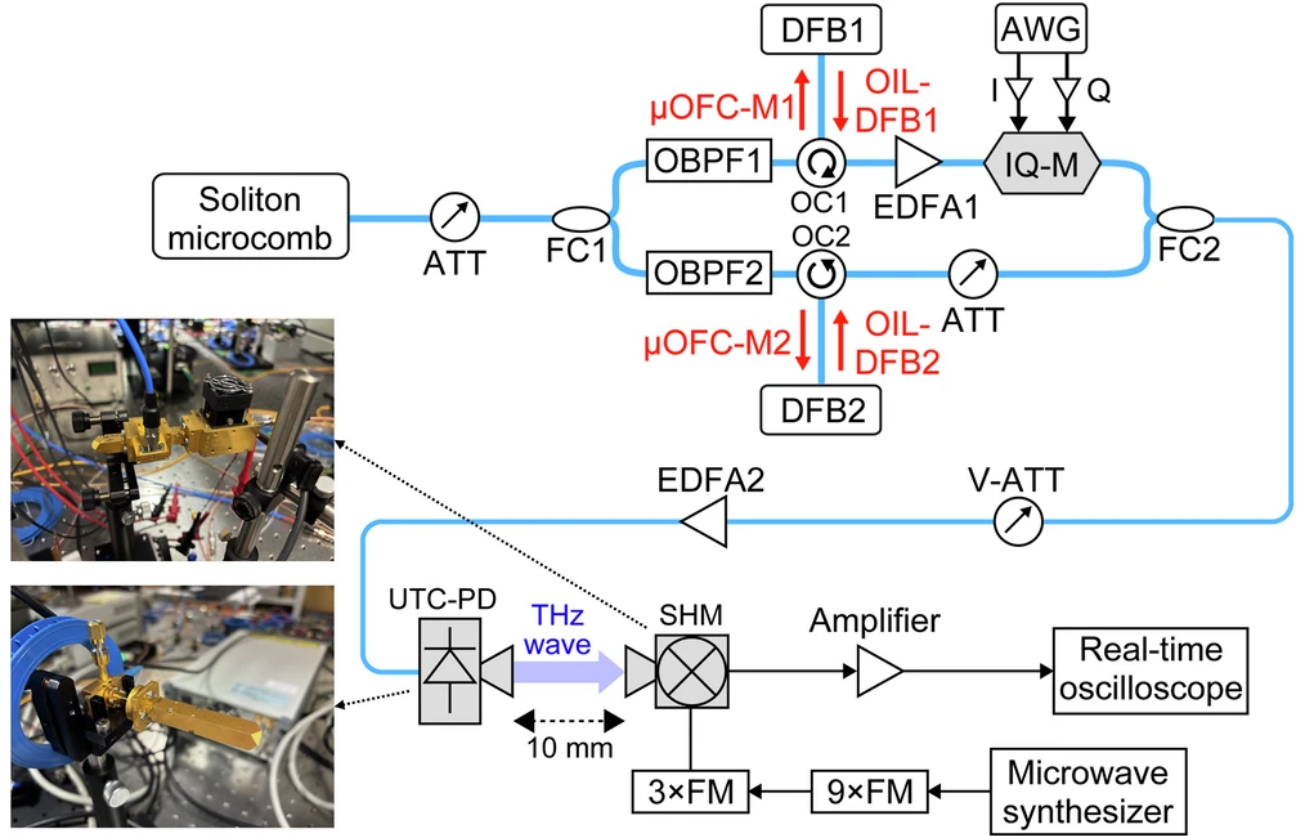


Fig. 11. Reproduced photonic wireless transmission set-up of Tokizane et al. A fibre-coupled SiN soliton microcomb feeds optical bandpass filters that select two adjacent comb lines; the lines optical-injection-lock two DFB lasers, one of which is IQ-modulated. The two beams are recombined and photomixed in a UTC-PD to produce an FSR-frequency wireless carrier (here 560 GHz). The same chain, with the microresonator FSR engineered to 100 GHz, supplies the drive for the on-chip launcher of Section 6.1. Reproduced from Tokizane et al. [4] under CC BY 4.0.

Three properties of this drive chain are specifically relevant to the plasmonic launcher of Section 6.1. First, the microcomb is *phase-coherent across all lines by construction*: the comb is a single physical mode-locked state, so the relative phase between any two lines is set by the soliton dynamics rather than by an external feedback loop, eliminating the dominant source of long-timescale phase drift in non-comb references. Second, the comb FSR is set by the SiN ring geometry and can be engineered with high precision; a 100-GHz FSR corresponds to a ring radius of order 230 μm , well within standard SiN-photonics processes. Third, the direct fibre-bonded packaging of Tokizane et al. [4] (Fig. 10) shrinks the entire drive source to ~ 5 mm, on the same physical scale as the half-adder chip; this makes the launcher, drive, and logic substrate genuinely co-integrable rather than coupled across a tabletop free-space system. The complete chain envisioned for the device is therefore: SiN soliton microcomb $\rightarrow$ bandpass filters $\rightarrow$ OIL-DFB pair $\rightarrow$ IQ modulator $\rightarrow$ UTC-PD $\rightarrow$ coplanar feedline $\rightarrow$ graphene patch antenna of Section 6.1.

6.4 Phase-locking precision: precedent from waveguide OPAs

The relative phase between the two photomixed tones must remain stable to within the $\pm 15^\circ$ budget derived in Section 5.3. Hirota et al. [8] have demonstrated 10.1-dB squeezed light from a broadband PPLN waveguide optical parametric amplifier by reducing the residual phase fluctuation of the phase-locking loop from 14 mrad to 9 mrad (Fig. 12). The reduction was achieved by replacing the conventional tap-the-working-signal scheme with a separate phase-detection OPA seeded by tapped probe and pump light *upstream* of the squeezer, so the working signal itself is never sampled and detection sensitivity can be raised independently of working-signal losses. The 9-mrad fluctuation they achieved is roughly thirty times tighter than our 260-mrad NOT budget; the same separate-arm topology, repurposed for the OIL chain of Section 6.3, gives a directly applicable phase-locking infrastructure.

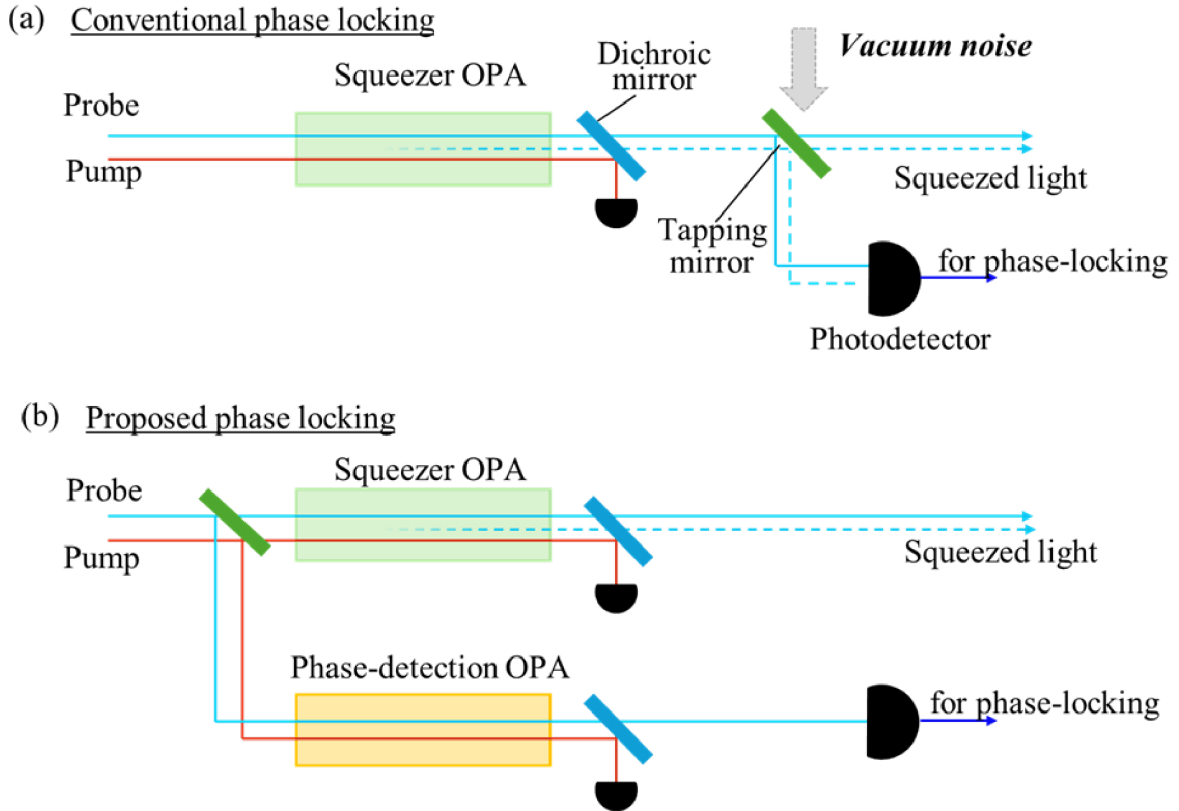


Fig. 12. Reproduced phase-locking topologies of Hirota et al. (a) Conventional method: a fraction of the working squeezed light is tapped after the squeezer OPA to generate the phase-error signal, coupling detection SNR directly to signal-path loss. (b) Proposed method: probe and pump are tapped upstream of the squeezer OPA and routed into a dedicated phase-detection OPA, so that the working signal is never sampled. Reproduced from Hirota et al. [8], *Opt. Express* 34, 7958 (2026), under the Optica Open Access Publishing Agreement.

7. Machine-learning-assisted design exploration

7.1 Design space

Four geometric and operational parameters dominate the device figure of merit: the cell pitch (range 0.5–2.0 μm), the top-gate voltage V_{Au1} (range -1.5 to -1.0 V), the bottom-gate voltage V_{Au2} (range -2.7 to -2.1 V), and the hBN gate-dielectric thickness h (range 5–15 nm). The natural single-number figure of merit is the NOT extinction ratio in dB.

7.2 Random Forest regression methodology

Following Yanagimoto et al. [14] and Haque et al. [3], we generate a synthetic dataset of design points by sweeping the four parameters uniformly over their ranges and evaluating the extinction ratio at each point under the analytic model of Section 5.3 augmented with a small simulation-noise term $\sum_{\text{sim}} = 0.3$ dB. The dataset is split 80/20 into training and test sets; a Random Forest regressor (200 trees, maximum depth 12) is trained and evaluated [5]. The dataset is small enough (~ 500 points) that the training cycle completes in seconds — orders of magnitude faster than a full-wave electromagnetic simulation sweep of the same coverage.

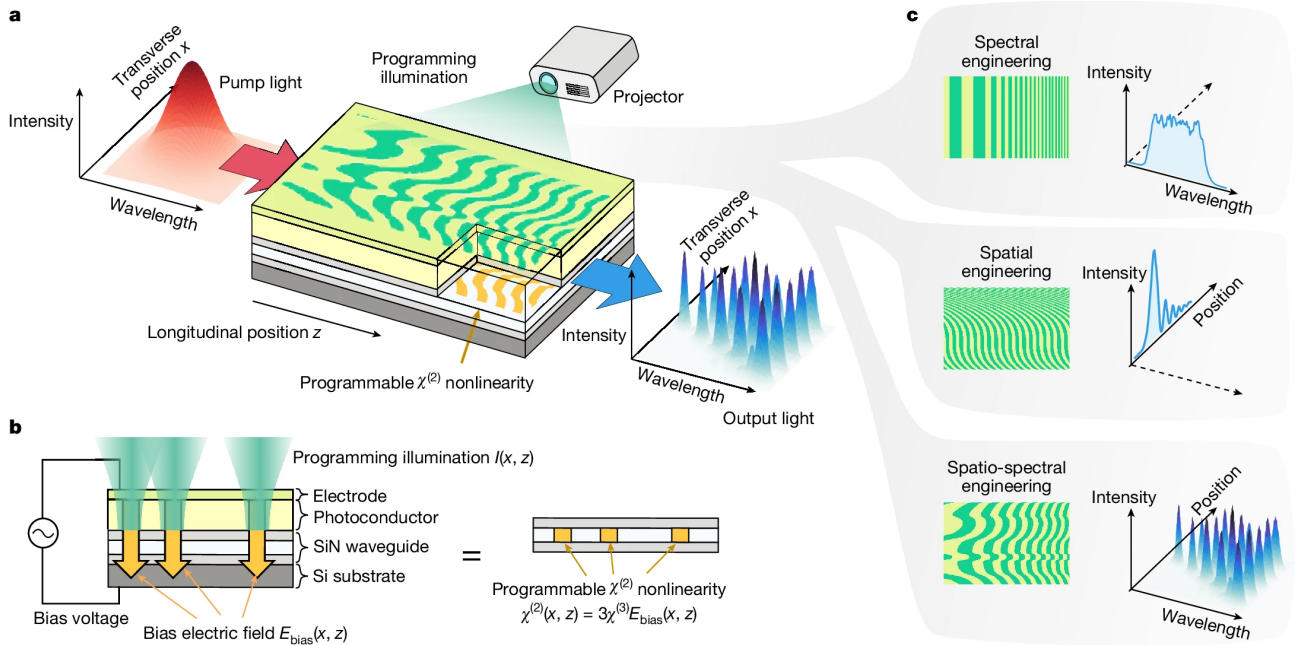


Fig. 13. Reproduced working principle of the programmable nonlinear photonic waveguide of Yanagimoto et al., which demonstrates lithography-free reconfiguration of a 2D function map under feedback control — the analogue paradigm for the machine-learning-driven design loop adopted here. Reproduced from Yanagimoto et al. [14], *Nature* 649, 330 (2026), under CC BY 4.0.

7.3 Predicted-vs-actual results

Figure 14 shows the predicted-vs-actual scatter on the test set. The model predicts the NOT extinction ratio with $R^2 = 0.94$ and $\text{RMSE} = 0.63$ dB, well below the 3 dB threshold at which the bit-error rate begins to degrade meaningfully. We emphasise that this dataset is derived from the analytic extinction model of Section 5.3 with additive simulation noise: production deployment of the surrogate for fabrication-grade design choices requires retraining on full-wave electromagnetic simulation output (e.g. CST Microwave Studio or COMSOL RF) so that the surrogate captures mode conversion, edge scattering, and antenna impedance mismatch — effects the analytic model cannot supply.

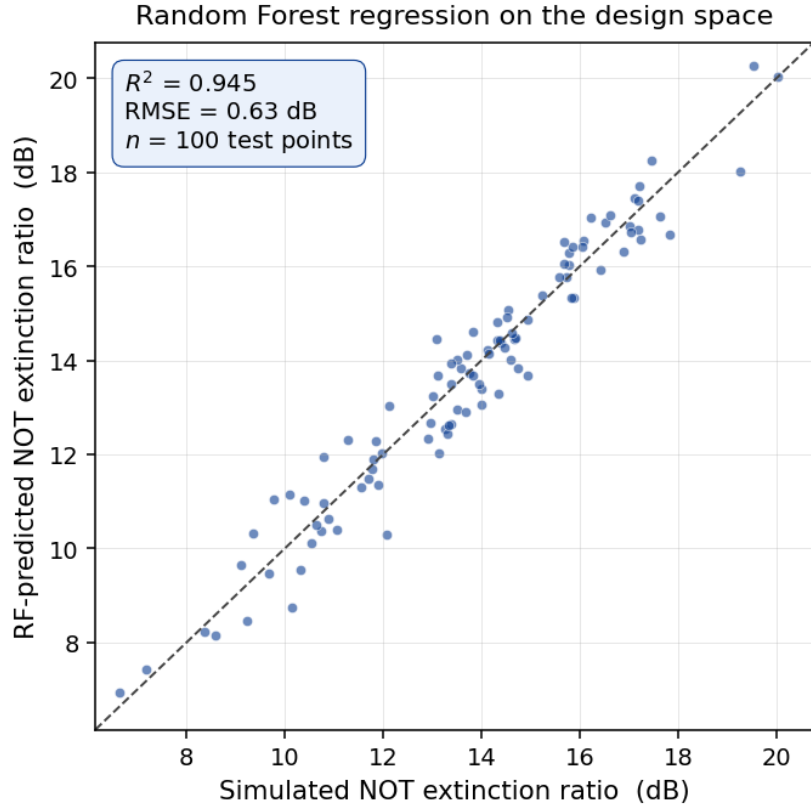


Fig. 14. Random Forest regression on the four-parameter design space (cell pitch, V_{Au1} , V_{Au2} , hBN gate-dielectric thickness), trained on synthetic simulation data. $R^2 = 0.94$, RMSE = 0.63 dB. Dashed line: ideal one-to-one prediction.

8. Limitations and paths forward

The 100-GHz operating point eliminates the cryogenic and femtosecond-phase-coherence constraints that have historically bounded plasmonic-logic demonstrations, but four engineering challenges remain. *Antenna and detector noise figure* at 100 GHz is set by the graphene patch quality factor and the detector NEP; the link-budget approach of Section 6.2 supplies the dominant design levers. *Cascading* with $L_p > 10 \mu\text{m}$ at $1\text{-}\mu\text{m}$ cell pitch accommodates a ten-deep gate cascade per re-launch stage; chip-scale designs will require an explicit re-launch every 8–10 cells — that is, within one plasmon coherence length. *Cross-talk and unintended scattering at lattice junctions* are a separate physics concern treated only by construction in the conceptual model above: edge scattering at active/CNP boundaries, mode conversion between odd and even stripline modes, and stray near-field coupling between adjacent rows can create leakage paths that degrade the NOT extinction ratio and pollute Sum/Carry read-out; their sizes are best bounded by full-wave electromagnetic simulation of the junction regions, which is the single highest-priority simulation task for a follow-on demonstrator. *Fabrication uniformity* across the 72-segment lattice will require either a per-segment CNP trim at test time or a real-time feedback loop that nulls offsets using the reference patch of Section 6.2 as the error signal; the machine-learning workflow of Section 7 supplies a fast model-in-the-loop calibration target.

9. Conclusions

We have presented a conceptual model of a 100-GHz, room-temperature half adder in which the bit is a graphene plasmon wavepacket on a sapphire substrate, driven by a microcomb-photomixing carrier chain in the geometry of Tokizane et al. [4]. The architecture inherits the field-coupled and Combo-gate vocabulary established in earlier conceptual work, but the operating point is chosen deliberately to eliminate the cryogenic and femtosecond-phase-coherence constraints of higher-frequency variants. The propagation length is $10\text{--}15 \mu\text{m}$, the phase tolerance is $\pm 125 \text{ nm}$ and $\pm 420 \text{ fs}$, and the intrinsic energy per bit is 10^{-17} J , all within the interior of what current photolithography and millimetre-wave instrumentation can deliver. Three precedents in adjacent fields give us particular confidence in the operating paradigm: Tokizane et al. [4] supply a chip-scale, fibre-bonded, phase-

coherent drive chain that scales to 100 GHz; Hirota et al. [8] show that the ~ 9 -mrad phase-locking precision we require for the NOT primitive is already routine in nonlinear-waveguide OPAs; and Kusyak et al. [7] have built the launcher + detector + reference monolithic platform on the same sapphire substrate adopted here. We hope this presentation is sufficient to invite full-wave electromagnetic simulation and fabrication trials by the graphene-plasmonics, microcomb-photonics, and 6G-antenna communities.

References

- [1] R. Furui, “Model of hybrid classical-quantum computing method,” available at Ryoji.info, Feb. 2019.
- [2] R. Furui, “Conceptual design of plasmon-generated programmable logic circuit, v2.0A,” available at Ryoji.info, Feb. 2024.
- [3] Md A. Haque, Md S. Akhter, I. Das, T. U. Zaman, J.-J. Tiang, N. S. S. Singh, A. D. Algarni, and A. A. Ateya, “Graphene based terahertz MIMO antenna with machine learning regression for 6G communications,” *Sci. Rep.* **16**, 3900 (2026); doi:10.1038/s41598-025-32487-9.
- [4] Y. Tokizane, H. Kishikawa, T. Kikuhara, M. Talara, Y. Makimoto, K. Yamaji, Y. Okamura, K. Nishimoto, E. Hase, I. Morohashi, A. Kanno, S. Hisatake, N. Kuse, T. Nagatsuma, and T. Yasui, “Beyond 350 GHz: Single-channel 112 Gbps photonic wireless transmission at 560 GHz using soliton microcombs,” *Commun. Eng.* **5**, 77 (2026); doi:10.1038/s44172-026-00659-8.
- [5] L. Breiman, “Random Forests,” *Machine Learning* **45**, 5–32, 2001.
- [6] K. Yoshioka, G. Bernard, T. Wakamura, et al., “On-chip transfer of ultrashort graphene plasmon wavepackets using terahertz electronics,” arXiv:2311.02821, 2023.
- [7] K. Kusyak, B. Schulte, T. Matsuyama, G. Kipp, H. M. Bretscher, M. W. Day, G. Meier, A. M. Potts, and J. W. McIver, “Monolithic optoelectronic circuit design for on-chip terahertz applications,” *APL Photonics* **10**, 076117, 25 Jul. 2025; doi:10.1063/5.0278183.
- [8] K. Hirota, T. Kashiwazaki, G. Ha, T. Yamashima, P. Jaturaphagorn, T. Suzuki, K. Takahashi, A. Kawasaki, A. Inoue, W. Asavanant, M. Endo, T. Umeki, and A. Furusawa, “Generation of 10-dB squeezed light from a broadband waveguide optical parametric amplifier with improved phase locking method,” *Opt. Express* **34**, 7958–7972, 9 Mar. 2026; doi:10.1364/OE.585323.
- [9] F. H. L. Koppens, D. E. Chang, F. J. García de Abajo, “Graphene plasmonics: a platform for strong light-matter interactions,” *Nano Lett.* **11**, 3370–3377, 2011.
- [10] M. Polini and F. H. L. Koppens, “Plasmons in graphene and other two-dimensional materials,” in *2D Materials for Nanophotonics*, Elsevier, 2020.
- [11] D. N. Basov, M. M. Fogler, F. J. García de Abajo, “Polaritons in van der Waals materials,” *Science* **354**, aag1992, 2016.
- [12] International Roadmap for Devices and Systems (IRDS), 2023 ed., IEEE.
- [13] D. A. B. Miller, “Attojoule optoelectronics for low-energy information processing and communications,” *J. Lightwave Technol.* **35**, 346–396, 2017.
- [14] R. Yanagimoto, B. A. Ash, M. M. Sohoni, M. M. Stein, Y. Zhao, F. Presutti, M. Jankowski, L. G. Wright, T. Onodera, and P. L. McMahon, “Programmable on-chip nonlinear photonics,” *Nature* **649**, 330–337, 8 Jan. 2026 (published online 8 Oct. 2025); doi:10.1038/s41586-025-09620-9.